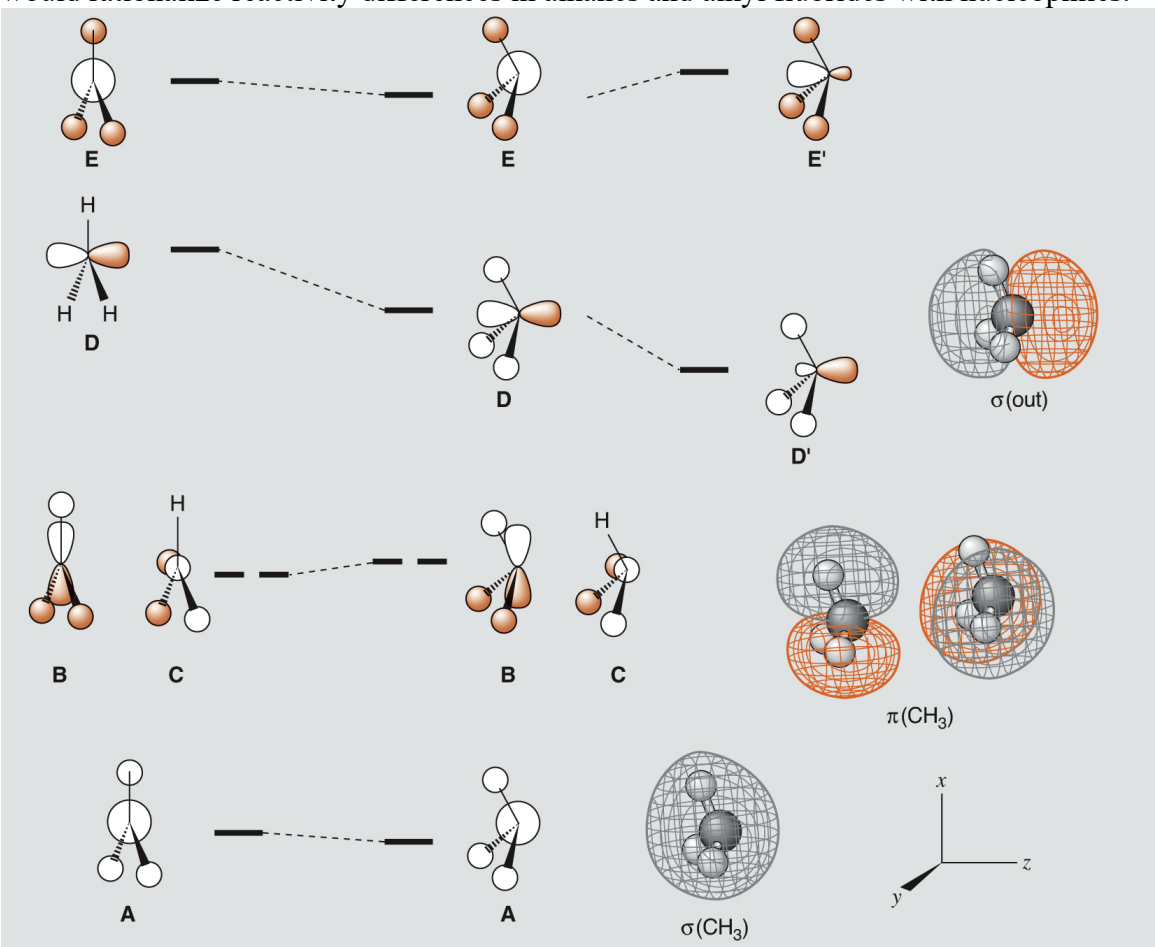
Organic Cume February 28, 2025

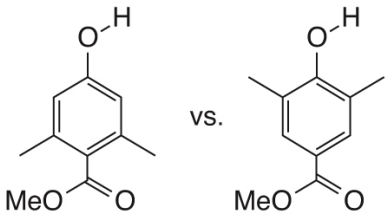
1. (Q21 of Ch 1, Anslyn) Draw mixing diagram using σ(out) orbitals that compare the formation of the C-C σ bond in ethane, created by the approach of two pyramidal methyl groups (see the diagram below), to the formation of the C-F σ bond in CH_3F. Use the sane energy axis for both plots. Highlight the differences and explain how this diagram would rationalize reactivity differences in alkanes and alkyl fluorides with nucleophiles.



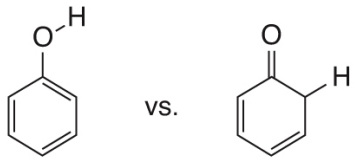
2. (Q20 of Ch 2, Anslyn) Predict the preferred conformation of fluoromethanol, FCH_2OH, around the C-O bond and briefly rationalize your choice.

3. (Q9 of Ch 5, Anslyn) In the following pairs of compounds, predict which is the stronger acid and explain why.

A.



B.



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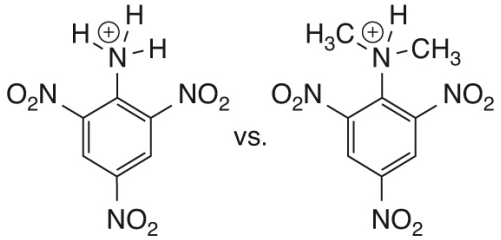
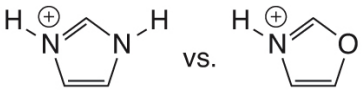
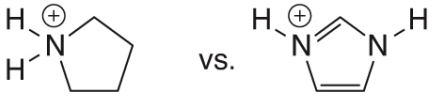


D.

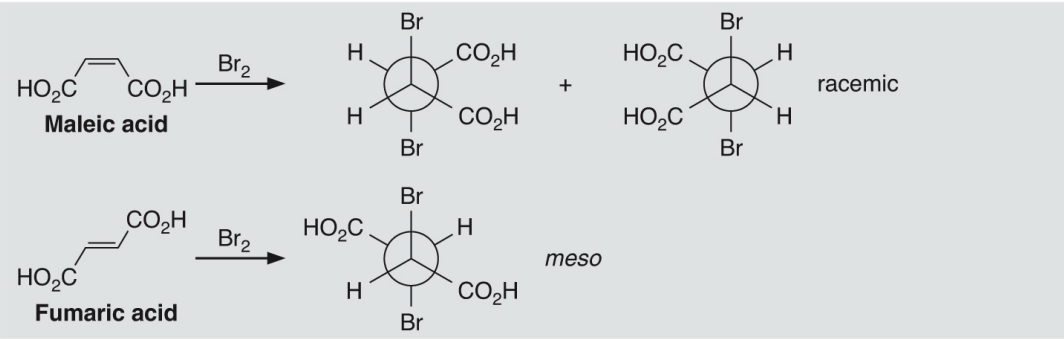


E.

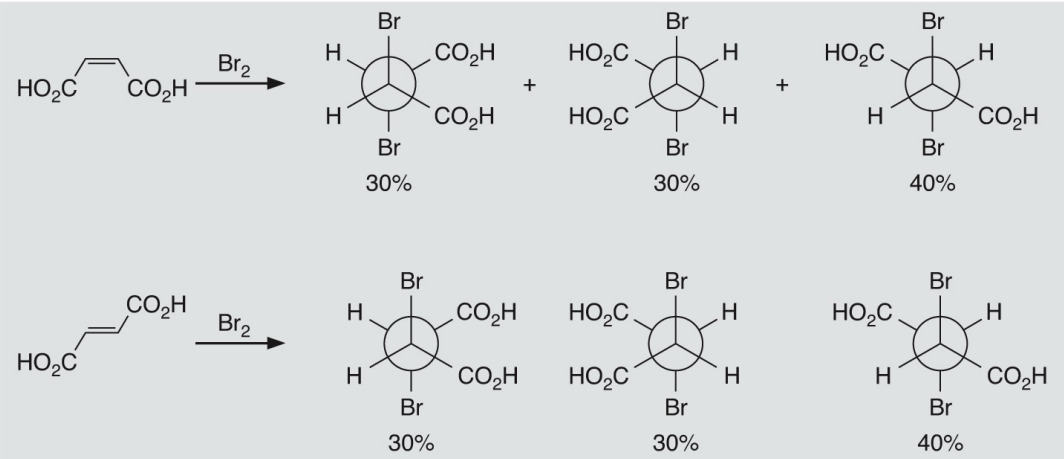


4. (Q22 of Ch 6, Anslyn) Define the following reactions as stereoselective and/or stereospecific, and if so, determine the percent stereoselectivity and/or stereospecificity. The products in A, D, E, and F are as shown. The product ratios in B and C are hypothetical for purpose of this question.

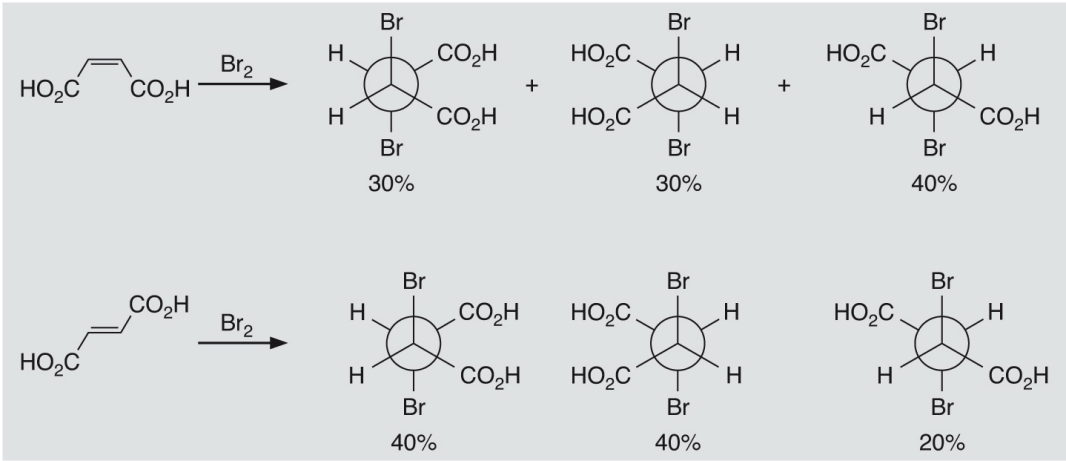
A.



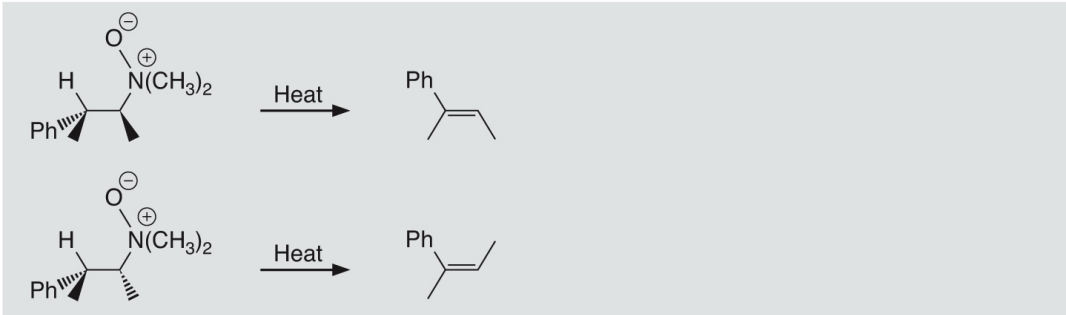
B.



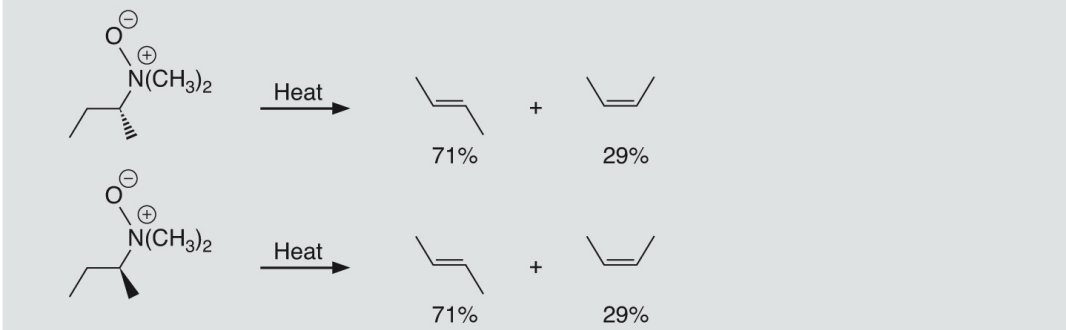
C.



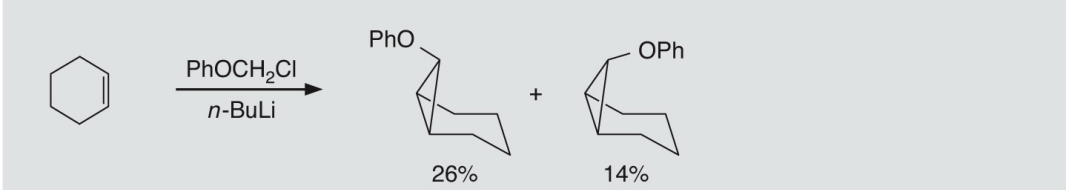
D.



E.



F.



5. (Q16 of Ch 7, Anslyn) The reaction of benzoyl chloride with ethanol and pyridine to form ethyl benzoate and pyridinium chloride can proceed either by nucleophilic attack of ethanol or pyridine in the first step. With nucleophilic attack by pyridine, an acyl-pyridinium species forms, so this kind of reaction amounts to nucleophilic catalysis by this added base. If ethanol attacks first, pyridine only acts to scavenge HCl formed in the reaction. Write these two possible mechanisms, and derive rate laws that distinguish these possibilities.